

MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

Computer Science and Engineering Department

MINOR PROJECT SYNOPSIS

B.Tech VII Semester, Batch (2023-2027)

Team ID	MNP059
Project Title	RAG-Based Language Agnostic Chatbot
Track	EdTech / Education Technology
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Project Guide	Ms. Prachi Gupta

1. Title of the Project

RAG-Based Language Agnostic Chatbot for Smart Query Response in Academic Institutions

A self-improving, intelligent Retrieval-Augmented Generation (RAG) chatbot designed to assist learners in gathering reliable academic and administrative information from institutional notices and circulars, with a primary focus on OCR-based chats for scanned noticeboard images, circular photos, and non-searchable image PDFs alongside digital documents. The chatbot can process multilingual and Hinglish queries and incorporates hybrid retrieval, factual verification, source citation [1], linguistic validation against language drift [2], and human assistance [4] in order to mitigate hallucination [1].

2. Problem Statement

The campus administration office gets many repetitive questions from students regarding scholarship forms, fee payment deadline notifications, exam schedules, eligibility conditions, and others [1], [4]. The official information is usually disseminated in English, but rather than clean digital text, it is predominantly published as scanned noticeboard images, physical circular photographs, and non-searchable image-based PDFs, alongside dynamic departmental web notifications. For students who are more fluent in Hindi or Hinglish, it becomes challenging to find and understand the necessary information. Moreover, a traditional chatbot which is based solely on naive RAG fails completely on scanned image notices, provides irrelevant context [1], fails to identify the exact notice or form ID numbers [3], and drifts away from the student's conversational language [2].

3. Why is the Particular Topic Chosen?

The topic was selected since students require precise and timely institutional data, which is dispersed across scanned notice images, physical circulars, and departmental website pages. The task incorporates a real-world problem related to academic administration [1], [4] as well as technologically interesting aspects such as OCR-based text extraction from noisy notice images, multi-lingual IR without parallel corpora [2], hybrid dense/sparse search [3], conversational re-formulation of queries [1], agent-based orchestration of workflows, fact-checking, and source-grounded generation [1], [2]. The proposed architecture directly tackles the failure modes of a naive RAG approach [1], [2] rather than treating retrieval and generation as one single process.

4. Objective(s) and Scope of the Project

Objectives:

- **Automate Institutional Data Ingestion & OCR Processing:** Continuously process institutional notifications, with primary capability to perform Optical Character Recognition (OCR) on scanned noticeboard images, circular photos, and raster PDFs alongside digital circulars and departmental notice-board content.

- **Enable Multilingual Querying:** Support English, Hindi, Marathi, Tamil, and Romanized Hinglish-style queries against English institutional documents and scanned circulars, bridging language barriers without parallel translation corpora [2], [4].
- **Improve Retrieval Accuracy:** Combine BGE-M3 dense retrieval with BM25 sparse lexical retrieval using Reciprocal Rank Fusion (RRF) and cross-encoder re-ranking to capture both semantic intent and exact circular codes [3].
- **Implement Self-Correction:** Use LangGraph to evaluate document relevance, reformulate failed queries, verify factual grounding, and retry generation when necessary, mitigating naive RAG retrieval bottlenecks [1].
- **Prevent Language Drift:** Detect the user's query language and validate that the final response remains in the same language, eliminating the documented ~27% multi-turn language drift failure [2].
- **Provide Verifiable Answers:** Attach structured source metadata (circular title, date, page/image source) and render clickable citation cards for institutional documents to ensure near-perfect factual attribution [1], [2].
- **Provide Safe Human Fallback:** Refuse to guess when reliable information is unavailable or when code-mixed queries cause ambiguity, escalating the query to administrative staff to maintain institutional trust [2], [4].
- **Support Auditability:** Log unresolved queries, timestamps, detected language, and related interaction data in a relational SQL database to streamline helpdesk workflows and knowledge base maintenance [1], [4].

Scope:

The prototype has functionalities that include OCR ingestion for scanned images and raster PDFs, digital PDF parsing, live noticeboard scraping, chunking with rich metadata, pre-processing for multi-lingual queries, hybrid search [3], LangGraph self-correcting decision making [1], contextualized response generation, factuality and language check [1], [2], citation generation [2], human escalation [4], and SQL based logging. The architecture includes an evaluation plan through RAGAS, multi-lingual consistency testing [2], and deflection of administrative workload [4]. This system is meant for institutional information and does not substitute administrative decision making.

5. Methodology, Process Description & Architecture

The system is organized as an automated ingestion and retrieval pipeline followed by a stateful LangGraph decision engine and a multilingual presentation layer.

Layer 1: Automated Ingestion and Multi-Modal Indexing Pipeline

- **Web Scraping Engine (BeautifulSoup / bs4)** periodically monitors active institutional notice boards and extracts announcements, publication timestamps, and administrative links.
- **OCR & Document Ingestion Engine (Tesseract OCR / EasyOCR & PyMuPDF)** serves as the primary gateway to process scanned noticeboard images, circular photos, and non-searchable image PDFs via pre-processing (binarization, deskewing) and OCR text extraction, alongside parsing digital PDF text, layout metadata, tables, and page boundaries.
- **Metadata-rich chunking** uses recursive semantic chunking with a 500-token window and 100-token overlap. Each chunk stores circular number, date, department, page/image index, and source URL metadata [3].

Layer 2: Hybrid Retrieval Engine

- **BGE-M3** provides multilingual dense vector retrieval in a shared 1024-dimensional semantic space, mapping regional student queries directly to English notice passages [2], [3].
- **BM25** provides sparse lexical retrieval for exact keywords, dates, circular identifiers, and form numbers, resolving dense retrieval blind spots [3].
- **Reciprocal Rank Fusion (RRF)**, with constant $k=60$, combines the dense and sparse rankings.
- **A cross-encoder** re-ranks the top 10 fused candidates and selects the final top 3 context passages [3].

Layer 3: Pre-Processing and Linguistic Router

- **A language detector** identifies the source language of the incoming student query [2].
- **For Romanized Hindi/Hinglish**, a normalization stage performs phonetic transliteration and extracts intent keywords to overcome code-mixing degradation [4].

- A **contextual query rewriter** resolves follow-up references using conversational state, such as interpreting 'When is its last date?' using the earlier scholarship context to avoid conversational context failure [1].

Layer 4: LangGraph Self-Corrective Decision Engine

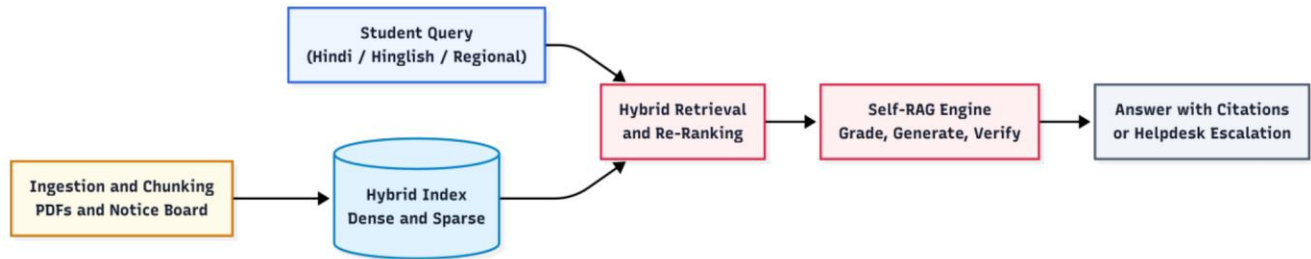
- **Node 1 – Document Relevance Grader:** checks whether retrieved passages contain the facts required to answer the query [1].
- **Node 2 – Query Reformulation and Web Fallback:** reformulates weak queries and can consult the live scraper when internal documents are insufficient [1].
- **Node 3 – Grounded Generation Engine:** generates answers strictly from verified passages and attaches source metadata to factual claims, ensuring zero hallucination [1], [2].
- **Node 4 – Hallucination and Factuality Grader:** compares the draft with source text and rejects unsourced dates, fees, or rules [1].
- **Node 5 – Language Consistency Validator:** ensures the response language matches the student's detected language while preserving citations, preventing language drift [2].

Decision and Escalation Rules

Decision Point	Condition Checked	Action Taken
Retrieval Assessment	Relevance score > 0.70 and key terms present [2], [3]	Proceed to grounded generation.
Ambiguity / Context Gap	Relevance score < 0.70 on initial retrieval [2]	Reformulate query and re-query vector store and live scraper [1].
Factuality Check	Generated facts lack direct grounding [1]	Reject draft and regenerate with stricter context bounds.
Language Verification	Draft language differs from detected query language [2]	Translate/alignment pass while preserving citation tags.
Escalation Trigger	Information remains absent after reformulation	Halt generation, show office contact card, and log query in SQL [2], [4].

The presented flowchart illustrates the general process of the proposed chatbot, ranging from query processing, OCR-based document and image ingestion, and hybrid retrieval to response generation and verification.

The flowchart also includes the self-correction mechanism and escalation to humans if there are no relevant or reliable resources found during the search [1], [4].



6. Hardware, Software, and Technology Stack

System Layer	Technology Selected	Engineering Rationale
Frontend UI	ReactJS, Tailwind CSS	Responsive multilingual chat interface, citation cards, and audio/voice input support.
Backend API	Python, Flask	Lightweight REST API routing requests between the UI and AI pipeline.
Document & Image Ingestion / OCR	Tesseract OCR / EasyOCR & PyMuPDF	Layout-aware digital PDF parsing and optical character recognition for scanned notices, photo circulars, and image PDFs.
Orchestration	LangGraph, LangChain	Stateful workflow for self-corrective loops [1], query rewriting, and language locking to prevent ~27% language drift [2].
Web Scraper	BeautifulSoup (bs4)	Crawls live departmental notice boards for recent campus updates.
Dense Embedder	BGE-M3 (BAAI)	Multilingual dense retrieval across regional queries and English circulars (97.6% top-k accuracy in Son et al. [3]).
Sparse Search	BM25 (Rank-BM25)	Exact matching for circular IDs, form numbers, dates, and keywords [3].
Vector Store	FAISS / ChromaDB	In-memory or persistent vector indexing for fast retrieval with sub-10 ms latency [3].
Relational DB	SQL (SQLite / PostgreSQL)	Session history, audit trails, feedback, and human helpdesk escalation queues [4].

7. Contribution Towards Society

This approach directly lowers administrative staff workload (targeting 30%–70% deflection of repetitive inquiries as reported in conversational AI literature [4]) while making official academic information more accessible. By prioritizing OCR-based chats, the system unlocks information previously trapped in physical noticeboard photographs and scanned image circulars. The multilingual and Hinglish normalization features facilitate access for students who do not feel comfortable communicating exclusively in English [4]. Verifiable source citations [1], [2] and confidence-based human fallback [2], [4] enhance the transparency and reliability of time-sensitive information such as deadlines, fees, forms, and examination schedules.

8. Schedule of the Project

Phase / Activity	Timeline
Requirement Analysis, Research Review & System Architecture	Weeks 1-2
OCR Ingestion Pipeline, Scanned Image/PDF Parsing & Metadata Chunking	Weeks 3-4
Hybrid Retrieval: BGE-M3, BM25, RRF & Re-Ranking	Weeks 5-7
Language Detection, Hinglish Normalization & Query Rewriting	Weeks 8-9
LangGraph Self-Corrective Decision Engine	Weeks 10-12
Grounded Generation, Factuality Checks & Language Validation	Weeks 13-14
Frontend Integration, Citations, SQL Logging & Testing	Weeks 15-16

9. Conclusion

The RAG-based Language Agnostic Chatbot introduces an agentic, self-correcting framework for institutional query resolution with specialized capability for OCR-based chats on scanned notice images and image PDFs. Moving beyond naive RAG paradigms [1], the architecture synthesizes multi-modal OCR ingestion, rich metadata indexing [3], dense-sparse hybrid retrieval [3], contextual query reformulation [1], LangGraph decision-making cycles [1], [2], grounded generation [1], factuality checking, language consistency validation [2], verifiable source citation [1], [2], and human fallback options [4]. The resulting system provides students with reliable, multilingual access to institutional notices without placing excessive burden on campus administrative offices [4].

10. References / Bibliography

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